

Programming Fundamentals

Assignment 3 - Part 4

Problem 1: Sum of All Values in a 2D Array

```
#include<iostream>
using namespace std;

int main()
{
    int rows, cols;
    cin >> rows >> cols;

    int marks[100][100];

    for(int i = 0; i < rows; i = i + 1)
    {
        for(int j = 0; j < cols; j = j + 1)
        {
            cin >> marks[i][j];
        }
    }

    int sum = 0;

    for(int i = 0; i < rows; i = i + 1)
    {
        for(int j = 0; j < cols; j = j + 1)
        {
            sum = sum + marks[i][j];
        }
    }

    cout << "Sum: " << sum << endl;

    return 0;
}
```

Problem 2: Subtract All Values from the First Element

```
#include<iostream>
using namespace std;

int main()
{
    int rows, cols;
    cin >> rows >> cols;

    int data[100][100];

    for(int i = 0; i < rows; i = i + 1)
    {
        for(int j = 0; j < cols; j = j + 1)
        {
            cin >> data[i][j];
        }
    }

    int result = data[0][0];

    for(int i = 0; i < rows; i = i + 1)
    {
        for(int j = 0; j < cols; j = j + 1)
        {
            if(i == 0 && j == 0)
            {
                continue;
            }

            result = result - data[i][j];
        }
    }

    cout << "Result: " << result << endl;

    return 0;
}
```

Problem 3: Area Calculator (Function Overloading)

```
#include<iostream>
using namespace std;

double calculateArea(double side)
{
    return side * side;
}

double calculateArea(double length, double width)
{
    return length * width;
}

int main()
{
    double side, length, width;

    cin >> side;
    cin >> length >> width;

    double squareArea = calculateArea(side);
    double rectangleArea = calculateArea(length, width);

    cout << "Area of Square: " << squareArea << endl;
    cout << "Area of Rectangle: " << rectangleArea << endl;

    return 0;
}
```

Problem 4: Bowling Average Calculator

```
#include<iostream>
using namespace std;

double getBowlingAverage(int runsConceded, int wicketsTaken)
{
    if(wicketsTaken == 0)
    {
        return 0.0;
    }

    return (double)runsConceded / wicketsTaken;
}

int main()
{
    int runsConceded, wicketsTaken;
    cin >> runsConceded >> wicketsTaken;

    double average = getBowlingAverage(runsConceded, wicketsTaken);

    cout << "Bowling Average: " << average << endl;

    return 0;
}
```

Problem 5: Vertical Word Printer

```
#include<iostream>
using namespace std;

void printVertical(char word[], int size)
{
    for(int i = 0; i < size; i = i + 1)
    {
        cout << word[i] << endl;
    }
}

int main()
{
    char word[100];
    cin >> word;

    int size = 0;

    while(word[size] != '\0')
    {
        size = size + 1;
    }

    printVertical(word, size);

    return 0;
}
```

Problem 6: Reverse a Word

```
#include<iostream>
using namespace std;

void reverseWord(char word[], int size)
{
    int start = 0;
    int end = size - 1;

    while(start < end)
    {
        char temp = word[start];
        word[start] = word[end];
        word[end] = temp;

        start = start + 1;
        end = end - 1;
    }
}

int main()
{
    char word[100];
    cin >> word;

    int size = 0;

    while(word[size] != '\0')
    {
        size = size + 1;
    }

    reverseWord(word, size);

    cout << "Reversed Word: " << word << endl;

    return 0;
}
```